HALMSTAD UNIVERSITY

Course Description

Robotic, 7.5 hp

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Table of Contents

1 Welcome! 3

1.1 Student's working environment 3

2 Course objectives and intended learning outcome 4

2.1 Course contents 4

3 Teaching formats 5

3.1 Lecture #1, Course introduction 5

3.2 Lecture #2, Image processing I 5

3.3 Lecture #3, Image processing II 5

3.4 Exercise #1, Introduction to the computer system and the camera 5

3.5 Lecture #4, computer system with software and hardware 5

3.6 Exercise #2: Multitasking and read data from laser scanner 5

3.7 Lab#1 (Examination) Find and show the position of an object 5

3.8 Lecture #5, Control theory 6

3.9 Exercise #3: Position controller 6

3.10 Lecture #6, Path planner and position feedback controller 6

3.11 Supervision 6

3.12 Lab #2 (Examination) Scan Match 6

3.13 Supervision 6

3.14 Supervision 6

3.15 Lab #3 (Examination) Kalman 6

3.16 Supervision 6

3.17 Final Examination 6

3.18 Specification of the recorded file 7

3.19 Written report 7

4 Grade and Examination 8

5 Final Contest 8

6 Time for examination 8

7 Preliminary schedule spring 2019 9

8 Course evaluation 9

9 List of components 12

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DT8022 – Course Description

1 Welcome!

The aim of this course description is to support your learning during this course. The goal is that this document will help you reach intended learning outcomes. The document contains a more detailed description of what you suppose to learn compare to the syllabus. It also describes your role as a student and what you suppose to do under and between different scheduled learning activities. Most of your time you will spend on this course is non-scheduled. The role of the teacher is to support you in your learning, but it is up to you how you learn. We also expect that you take responsibility for your learning, meaning that you plan your studies so that you reach the course objectives in time. Some of the learning activities are done together with one or more of your fellow students, e.g. project and exercises. This requires that you are well prepared before these learning activities so you have something to contribute with. Studies show that you actually learn a lot from your fellow students and these learning activities are designed based on this.

There are some requirements to pass this course. This document tells you what these requirements are and how they will be assessed. It is important for us that you feel that your examination has been correctly assessed and that it is according to current regulations.

The course has a homepage where you can find most of the material and information given in this document and some additional material see:

https://hh.blackboard.com/

Contact information

Teacher/ Course examiner	Tommy Salomonsson and Björn Åstrand
Mail	tommy.salomonsson@hh.se and bjorn.astrand@hh.se

Important! **Mark all mail with "DT8022" in the subject field. Please keep track on all history in mailing conversation. Use "reply"!**
You shall also use your university mail address in conversation. According to University regulations you should regularly check your email box and read relevant messages contained therein.

1.1 Student's working environment

"Creating a good working environment is something which is collectively of importance for management, employees and students all of whom have a responsibility to participate in the process". To read more about Halmstad University´s program for systemic working environment initiatives, use following link:
http://www.hh.se/english/education/registeredstudent/commonregulations/workingenvironment.2369_en.html

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2 Course objectives and intended learning outcome

The goal of the course is for students to gain knowledge on how to design and control an autonomous mobile robot.

The course is included as an elective course in the Master's Program in Embedded and Intelligent Systems 120 credits. The course is a mandatory course in the program Computer Science and Engineering, 300 credits, with specialization intelligent systems.

Following successful completion of the course, the student should be able to:

Knowledge and understanding

- Explain different methods for global and local path planning works and when different methods are appropriate.
- Describe how motion control of a mobile robot is done by using a kinematic model.
- Explain how low level control of actuators is done using position and speed and acceleration proles.

Skills and abilities

- Design and implement an autonomous robot, including mechanics, computer software and hardware.
- Design and implement a complete navigation system including perception, localization, path planning and motion control.

Judgement and approach

- Judge whether a particular solution is appropriate for a given task, and suggest alternative solutions if appropriate, both on system and sub-system level.

2.1 Course contents

The lectures present different methods for design and control of an autonomous mobile robot; techniques for path planning and motion control (kinematic model, control of position, velocity and acceleration); perception (cameras, range sensors, encoders); obstacle avoidance strategies; navigation architectures.

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3 Teaching formats

Lectures introduce concepts and methods, simulator based exercises serve to deepening the understanding of theory and develop practical skills needed for the project. An evaluated team based project completes the practical part of the course. Project implementations will run on a real robot and shall solve a predefined task.

Some lectures are given, but the major part of the teaching is laboratory exercises and project work including supervision. The students working group of 2 or 3 students per project.

Teaching language will be in English.

The rest of this chapter is a compilation of the different scheduled teaching and learning activities in this course.

3.1 Lecture #1, Course introduction

The course starts with a course introduction. We will also walk-through this course description, and you can ask questions if there are some unclarities.

The project is that an autonomous mobile robot is built and programmed by the students. The robot is builded with Lego bricks, sensors, actuators, color-camera and a microcontroller. The microcontroller is programmed making the robots totally autonomous during the examination. The students work in groups of two or three.

3.2 Lecture #2, Image processing I

On this lecture, we will talk about how to get a binary image from a color image and introduce different color spaces.

3.3 Lecture #3, Image processing II

On this lecture, we will talk about how to detect and classify objects seen by the color camera. We also talk about different morphological operations on binary images and methods to calculate center of gravity of an object.

3.4 Exercise #1, Introduction to the computer system and the camera

This exercise is an introduction to the tool used for programming the computer board used for controlling the robot, i.e. process sensor data and control actuators.

3.5 Lecture #4, computer system with software and hardware

On this lecture, we will talk about how the computer is connected to sensors and actuators. How sensor values can be read and how actuators can be controlled. We will also talk about how code can be executed with different sample time.

3.6 Exercise #2: Multitasking and read data from laser scanner

On this exercise, we will study different problems with real-time and multitasking. We will also read data from the laser scanner.

3.7 Lab#1 (Examination) Find and show the position of an object

This examination is about a program that use the camera as a sensor to find an object. Write a program that will print the distance in pixels between the center of an object and the center of the image. This should be done with 100ms sample time. The student can choose any object.

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3.8 Lecture #5, Control theory

On this lecture, we will talk about speed and position controller of a dc-motor.

3.9 Exercise #3: Position controller

On this exercise, we will learn how to control the dc motors with a velocity profile.

3.10 Lecture #6, Path planner and position feedback controller

On this lecture, we will talk about the different kinematic models (ackermann steering, differential drive). How to use different sensors for localization (encoders, cameras, laser-scanners) Odometry and scan match. This is a preparation for the laboration 3.

3.11 Lecture #7, Realtime, Scheduling and multitask

On this lecture, we will talk about the problem with real time and one solution about how to update the estimated positions.

3.12 Supervision

This is scheduled time for getting help with the projects and asks questions. There is limited time for supervision outside these occasions.

3.13 Lab #2 (Examination) Scan Match

Show that a position of a laser-scanner inside a paper box can be calculated with a scan match algorithm. The time, estimated position [mm] and uncertainty [cov matrix] should be printed out with 1sec sample time.

3.14 Supervision

3.15 Supervision

3.16 Lab #3 (Examination) Kalman

Show that the robot can estimate it's position with Kalman filter by help of the encoders and by the laser-scanner. The time, estimated position and uncertainty should be printed out and saved to file with 100ms sample time. The student need to:

- Build and program the robot so it moves from A to B according to a velocity profile.
- Save the estimated positions and the uncertainty from the Odometry, Scan Match and Kalman to file for evaluation, se section below (Specification of the recorded file).

3.17 Supervision

3.18 Final Examination

The examination is to show that the robot solves a predefined task autonomously in a given time frame.

The predefined task is to fetch 2 blue boxes and leave them in a given area within 3 minutes. The given area is 300x300 mm at the center of the sidewall. From start there will be within the arena four blue boxes randomly placed. There can also be other boxes with other colors inside the track. The task is assessed given following rules/rights:

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- The task shall be performed fully autonomously. No interaction of a human is allowed after start. This includes remote control by human, picking up lost/dropped lego bricks ect.
- The camera must be used for detection and classification of the object.
- The LIDAR sensor together with the encoders must be used for localization.
- The estimated positions and uncertainty should be recorded and saved to file for evaluation, se section below.
- Maximum two trials for each robot team.
- Within 20 seconds from start the team has the possibility for a restart, e.g. to handle situations there the system fail to start. This can be done maximum two times for each trial. No restart is allowed after 20 seconds.
- The starting position of the robot is at the center of the sidewall.
- If one or more wrong colored box picked, then failed.
- All students within the group must be physical participate during the examination. No remote is allowed by Zoom or anything like that.

To pass this examination following rules must also be followed:

- Not using any other parts besides those listed in section 12.
- That no Lego bricks are glued or manipulated (modified in any way).
- That the maximum size of the robot is 400x250x1000 (LxWxH) mm at start.
- Electrical wires and passive electrical components are allowed (resistors and capacitors).
- Tape is allowed to attach sensors and actuators.
- None of the electrical components (e.g. wires) or tapes is allowed to use for enhancing the mechanical construction.
- It's not allowed to drop objects on the arena intentionally.

3.19 Specification of the recorded file

Tre files should be created from Odometry, Scan-match and Kalman calculations. These recorded files should be text files that can be opened and readable by Matlab. Separate each measurement by lines and values with spaces in the order below.

Time [s]	X [mm]	Y [mm]	θ [rad]	σ_{11}	σ_{12}	…	σ_{33}

3.20 Written report

The written report shall be written according to the guidelines given in "**Guidelines for writing technical reports**" see appendix. Send, by e-mail, the report in pdf-format to the course examiner bjorn.astrand@hh.se. Do **not** forget to use the course code in the subject field, i.e. DT8022. The following part shall be included in the report:

- The strategy (How, in what order and so on about fetching the boxes).
- The model of the robot (How is it controlled).
- Image processing (How to find and classifying object eq with machine learning).
- How to use the camera as a sensor (What to do when you have found the box etc.).
- How to use the LIDAR, encoder (Localization)
- Path planning (witch algorithm eq Spline)
- The controller (the software, your program, the hardware, etc.).
- The actuators (DC-motors, Servo).

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- The manipulator (How to fetch the boxes).
- Detect obstacles (sensor used, method).
- The gearbox (design, selection of ratio).

4 Grade and Examination

The requirement to pass this course is:

- Show that the robot solves the predefined task stated in section final examination.

To get grade 4 the additional requirements is:

- Show that the robot solves the predefined task stated in section final examination.
- Pass the tree laborations, Camera, Cox and Kalman, explained above.
- Pass the extra task with Machine Learning **OR** the extra task with Spline.
- Hand in a written report that describes the complete robotic system following the guidance given in section 3.19.
- Participate with a well working and robust robot in the final contest.

To get grade 5 the additional requirements is:

- Show that the robot solves the predefined task stated in section final examination.
- Pass the tree laborations, Camera, Cox and Kalman, explained above.
- Pass the extra task with Machine Learning **AND** the extra task with Spline.
- Participate with a well working and robust robot in the final contest.
- Hand in a written report that describes the complete robotic system following the guidance given in section 3.19.
- Within the report, there should also be a section about analyzing/discussion. Describe how well the robot perform the task and how it can be approved.

The written report on the system has to be well written. Like a scientific paper, contain relevant information and well explained results. You will get feedback **one time only** on the written report.

For more information read the grading criteria's.

5 Final Contest

Fetch as many boxes you can within 3min. The one that had collected the most blue boxes without any "wrong" colored after 3min will win. Stealing from each other is an option not a must. Leave a box with wrong color in the opponent's "home" is also an option. The winner of the competition doesn't need to disassemble the robot after it is done and Martin Cooney will have some questions about how you have implemented it.

6 Time for examination

The course ends in week 21 with the first examination occasion in week 21 for the final exam and the possibility of retaking it in week 22. Last day to hand in the report is in week 23.

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The extra task for higher grade like machine learning and or spline will be done at the same time as the final exam.

Last occasion to show the robot work is just before the start of the autumn semester (the day is not decided yet). For those who will do the examination after summer the last occasion to hand in the report is one week after the examination. Except these days there are no possibility too do an exam.

7 Preliminary schedule spring 2025

Preliminarily schedule – always check the schedule for the current week on University homepage.

	Lecture	Supervision	Exercise	Examination
w.14	L#1 Course description, Overview L#2 Image processing I L#3 Image processing II			
w. 15	L#4 Computer system and Multitasking		Exercise#1 Introduction to the computer system Grabbing images I	
w. 16	L#5 Control the motors and Communication.		Exercise #2 Laser scanner, multitasking	Examination I Track object
w. 17	L#6 Connect everything together		Exercise #3 Control the motors	
w. 18	L#7 Spline and Guest lecture about Machine Learning	Supervision and reexam nbr1 EI		
w. 19	L#8 Realtime	Supervision and reexam nbr2 EI		Examination II Scan Match
w. 20		Supervision and reexam nbr1 EII		Examination III Kalman
w. 21		Supervision, reexam nbr2 EII reexam nbr1 E1II		
w. 22				Final examination and reexam nbr2 EIII
w. 23				Reexam nbr1

8 Course evaluation

At the end of the course, there will be a course evaluation. Here you have the possibility to give feedback on the course, hopefully in a constructive way, so it can be used as guidance for future development and planning of the course. However, giving feedback on the course is not restricted to this occasion. You can always discuss various matters of the course with the examiner.

The course evaluation is done online and a mail containing instructions will be sent to you at the end of the course.

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Guidelines for writing technical reports

Ola Bengtsson

Abstract

You should summarize the most important results of this exercise and write it in the abstract. The abstract should also include a short description of which area that is covered by the exercise and why this is important. The abstract should be 15 sentences long approximately.

1. Introduction

You should describe the area covered in the exercise. The description should be such that any person with appropriate technical knowledge should understand where and when to use the theory and methods described in this report. You might want to present, and briefly describe, work related to the work presented in the next section, Theory / Method. This means that you e.g. should refer to literature that gives a deeper presentation of the related work [Person and Person 2000]. More information can be found in [Person et al. 2001]. You should e.g. add the limitations in the experiments and under what assumptions the experiments where done. The introduction part should be one half to one page long.

To make the report consistent with scientific reports, the introduction part should also include a short description of the rest of the report, i.e. a brief description on what follows in the report.

2. Theory / Method

The experiments should be replicable, which means you should describe it in such way that any person with appropriate technical knowledge should be able to replicate your experiments by following your description in Section 2 and 3. This means that Section 2 should include all necessary theory, e.g. mathematical equations, references to necessary papers, illustrative figures etc. (Sections 2 and 3 should be the main core of the report.)

It is important to get a good structure of Section 2, why the use of sub-sections (i.e. Section 2.1, 2.2 and Section 2.2.1, 2.2.2 etc.) might be appropriate.

3. Experiments / results

You should describe the conducted experiments in such way that any person with appropriate technical knowledge can repeat the experiments. It is important to clearly explain the purpose of each experiment, show the results in illustrative ways and explain the results to the reader. In the case of many experiments, describe each experiment in a separate sub-section, i.e. Section 3.1, 3.2, 3.3, …, 3.N.

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4. Conclusions and further improvements

You should make a clear conclusion on the exercise, what you have learnt etc. It is also appreciated if you write suggestion on how to improve the exercise for another year.

References

[Person and Person 2000] Person, O.; Person, T. (2000), Title of the paper that I am referring to, Which Journal or Proceedings the Paper was Published, In which town the paper was published, What year the paper was published, In what volume the paper was published, In which pages the paper was published.

[Person et al. 2001] Person, O.; Person, T.; Person, T. Person, F. (2001), Title of the paper that I am referring to, Which Journal or Proceedings the Paper was Published, In which town the paper was published, What year the paper was published, In what volume the paper was published, In which pages the paper was published.

Send, by e-mail, the exercise report in pdf-format to the course examiner bjorn.astrand@hh.se. Don't forget to use the course code in the subject field, i.e. DT8022.

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9 List of components

Lego	
Bulks	no
16*1	20
12*1	20
10*1	10
8*1	30
6*1	25
4*1	50
2*1	50
Plates	no
10*6	8
6*4	8
4*4	8
10*2	6
8*2	30
6*2	15
4*2	40
3*2	20
2*2	20
10*1	20
8*1	30
6*1	30
4*1	40
3*1	40
2*1	50

Wheels and shafts	
Wheel	no
Large	4
Medium	4
Small	4
Wheel with ring	10
Wheel without ring	4
Shaft	
12	10
10	10
8	10
6	20
4	20
3	10
2	10

Actuators and sensors	
Actuators	no
DC-motor	2
Position servo	2
Sensors	no
Micro switch large	2

Cogwheels and related components	
Cogwheel	no
40	10
24	30
16	25
8	20
Wide chain	40
Narrow chain	80
Differential	1
Cogwheel diff.	4
Cardan joint	2
Threaded bar	2
Nut	2
Cogged bar	10

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